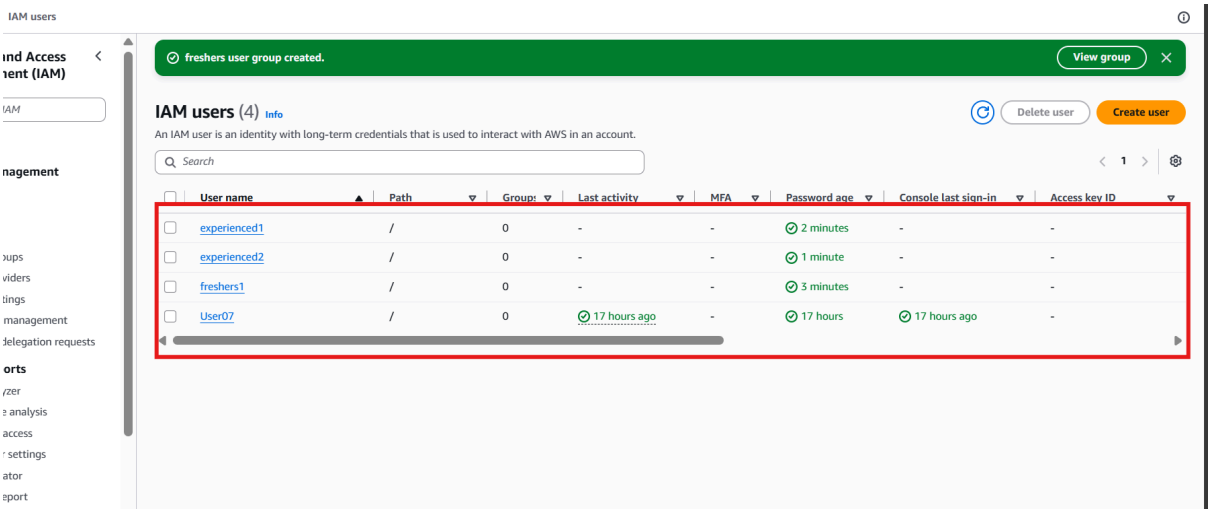
Core Concept: EC2 + EBS Practical

AWS Service → Actions → Policies → IAM User/Group

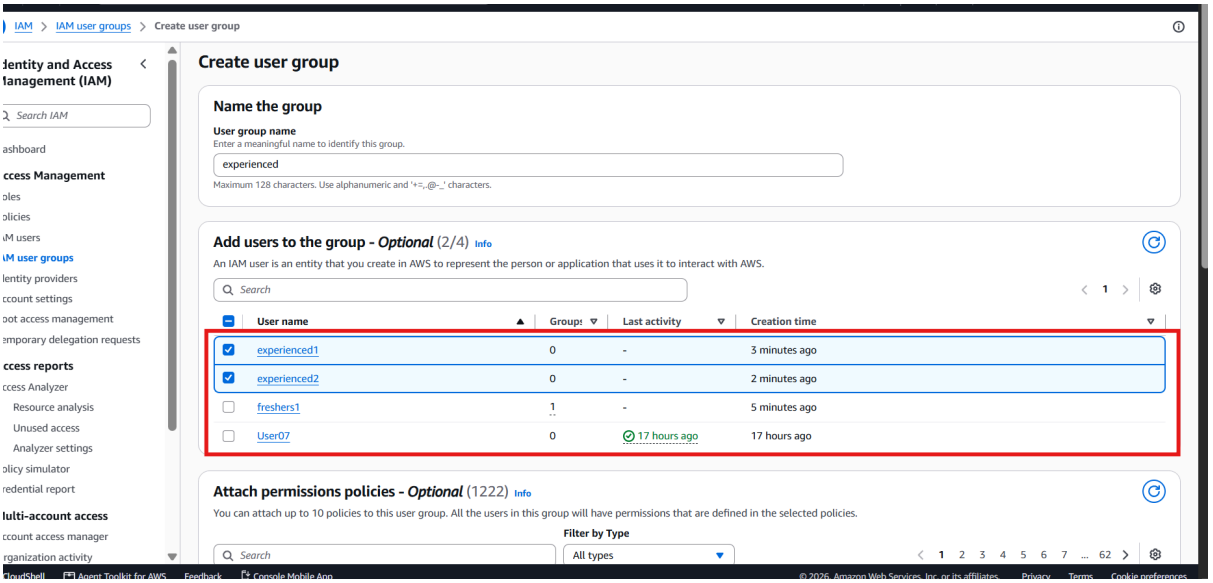
For EC2:

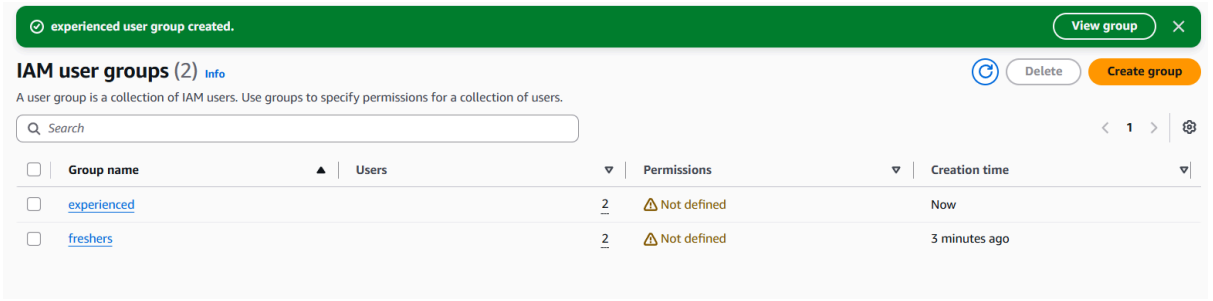
- **Actions** are individual permissions, e.g. `ec2:AttachVolume`, `ec2:DetachVolume`, `ec2:RunInstances`.
- These actions can be included in **IAM policies**.
- AWS provides many **AWS-managed policies** containing collections of these actions, such as `AmazonEC2FullAccess` and `AmazonEC2ReadOnlyAccess`.
- You **attach the policy** to an IAM user/group, and that determines what actions they are allowed to perform.

Step 1: Create four users with the intend of separating into two groups.

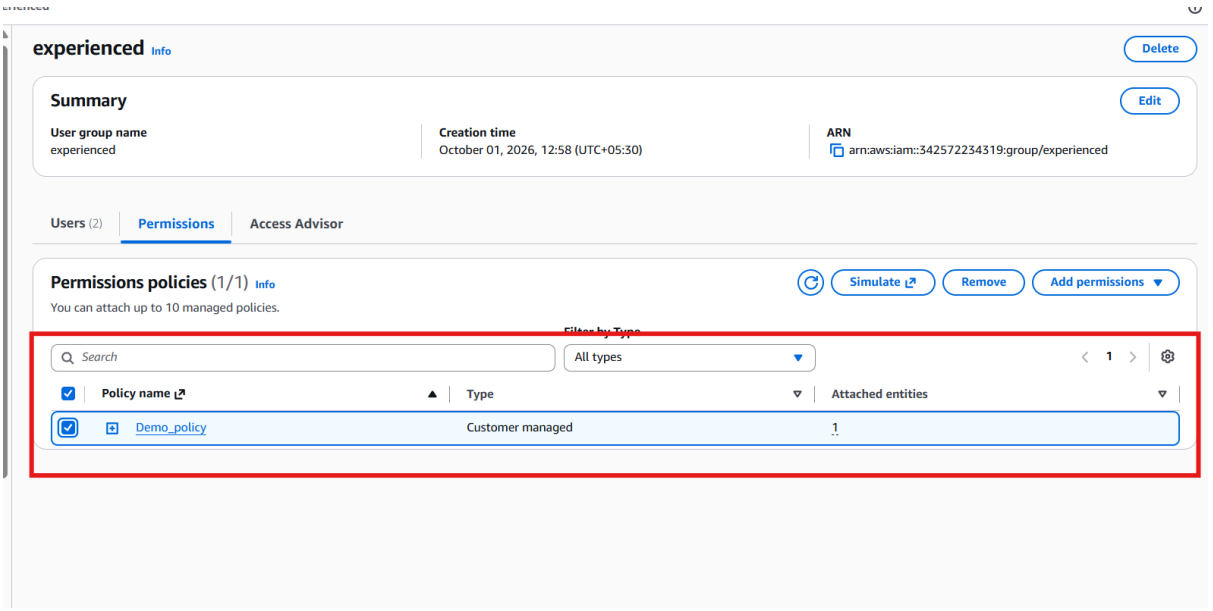


Step 2: Create two user groups as “fresher” and “expereinced”.

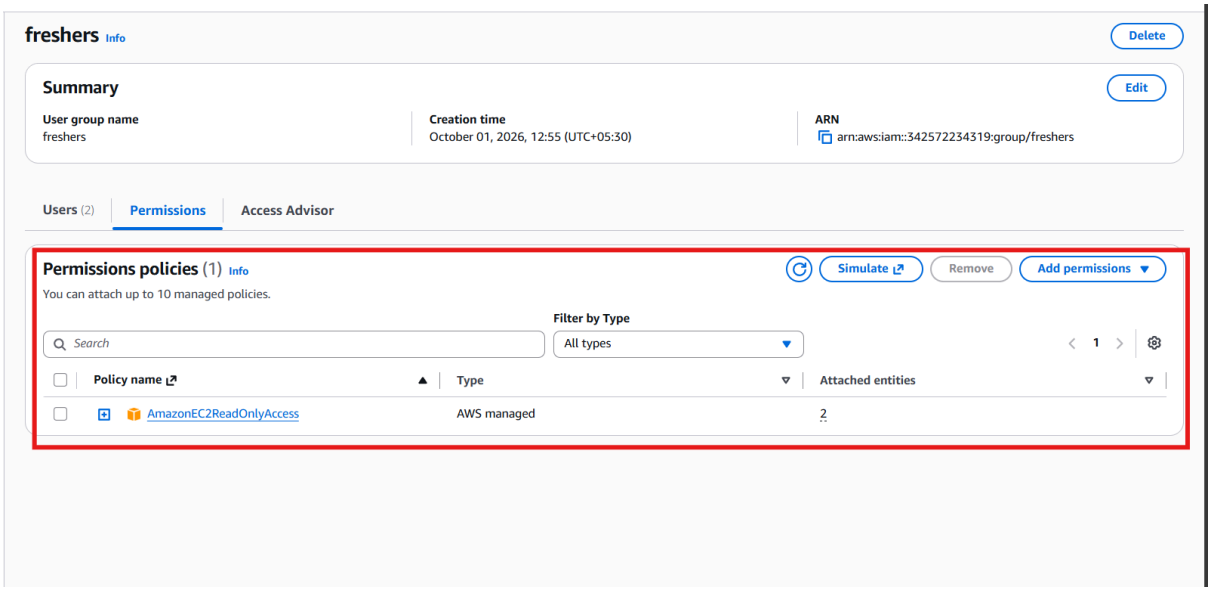




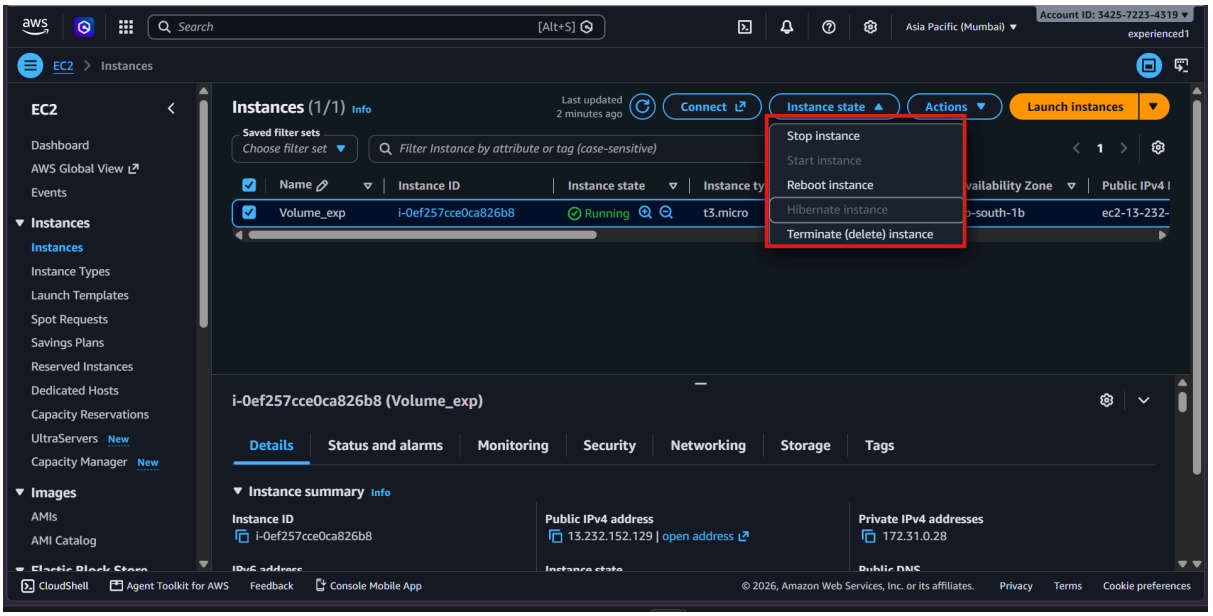
Step 3: Attach the policy of Stop, Start and Terminate to the Experieneced Group.

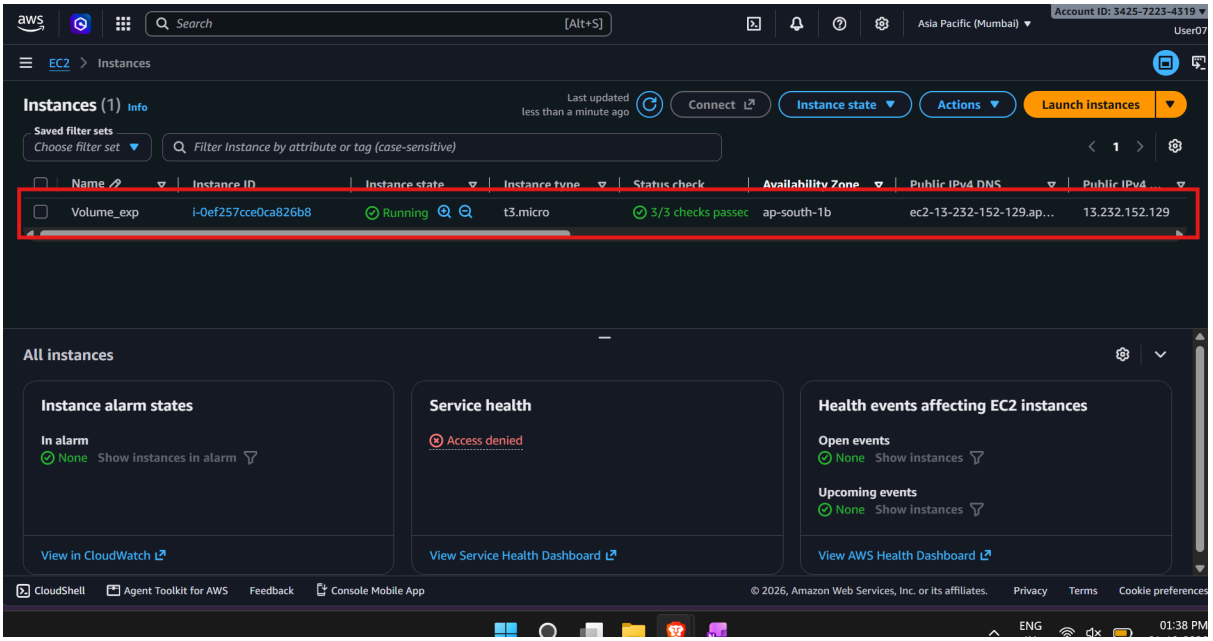


Step 4: Attach the policy of EC2ReadOnlyInstance to the fresher group.

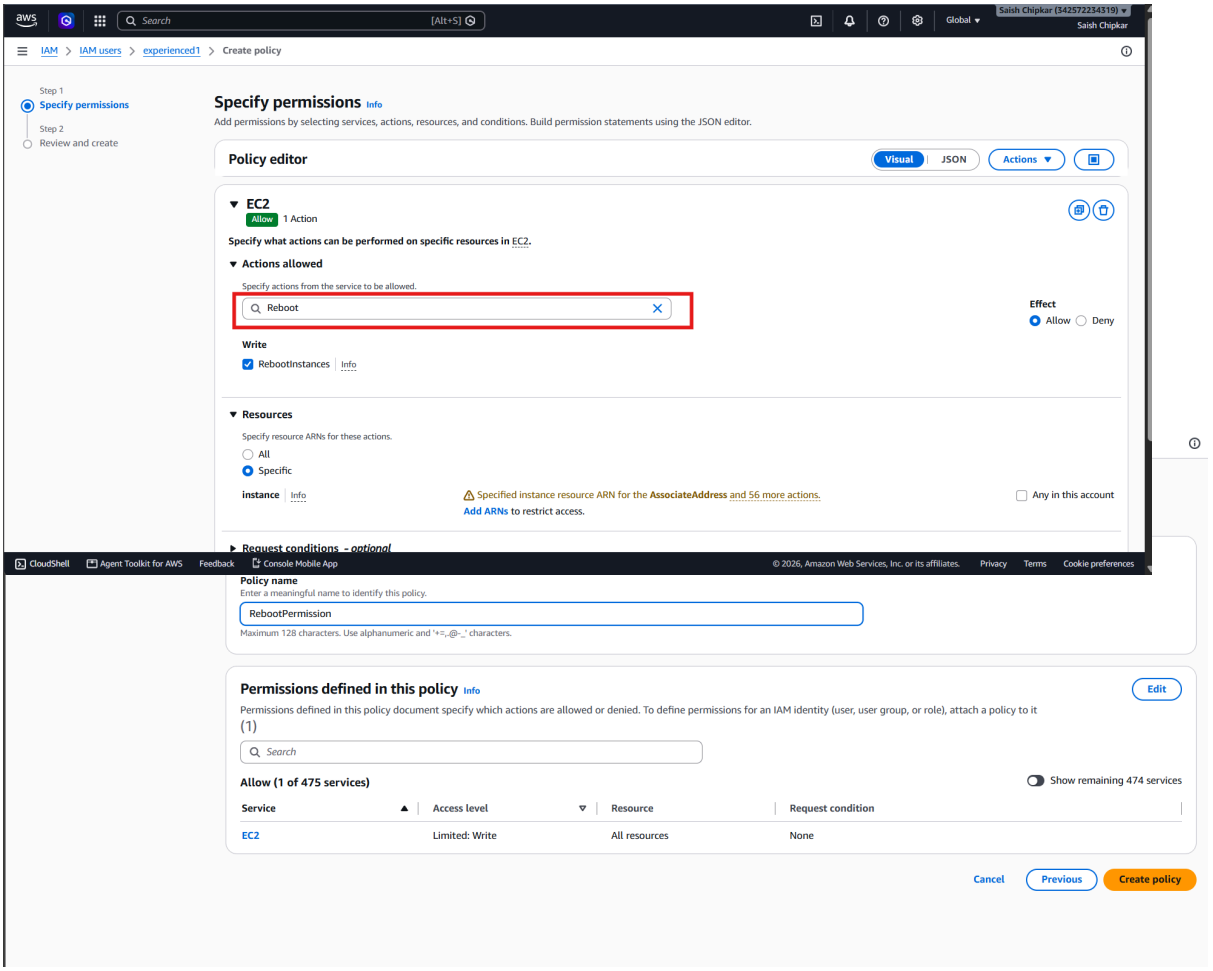


Step 5: Check the actions from both Freshers and Expereinced IAM accounts.

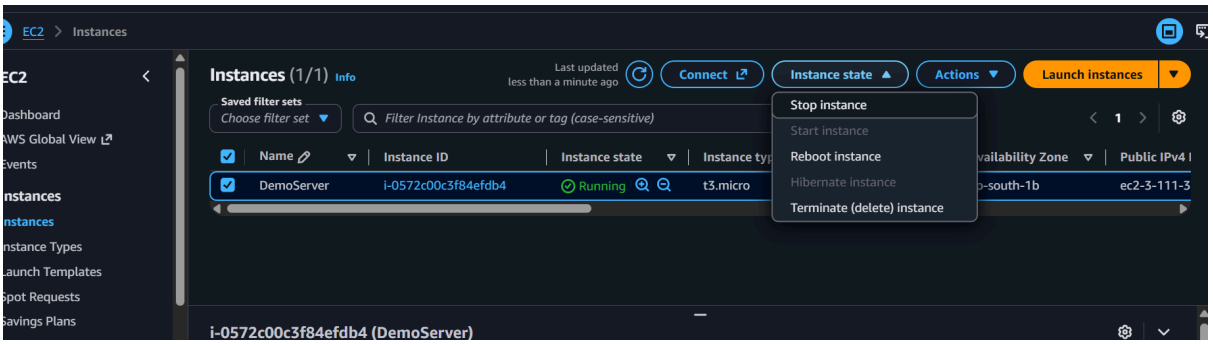




Step 6: Provide Reboot action permission to the expereinced1 IAM user using the Inline policy.

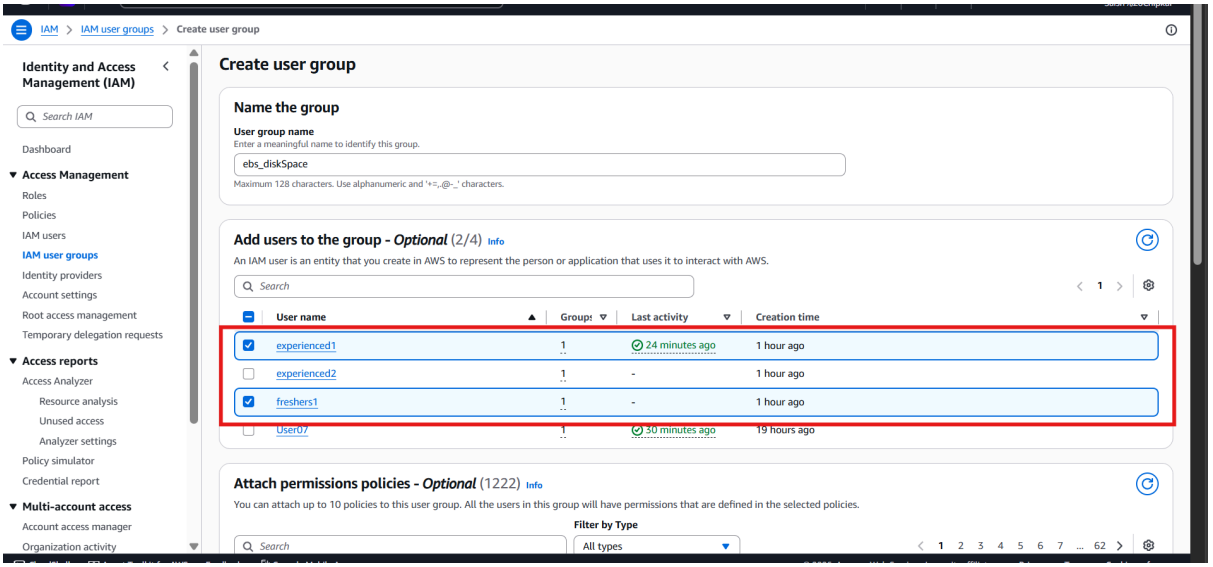


Expereinced1 Console:



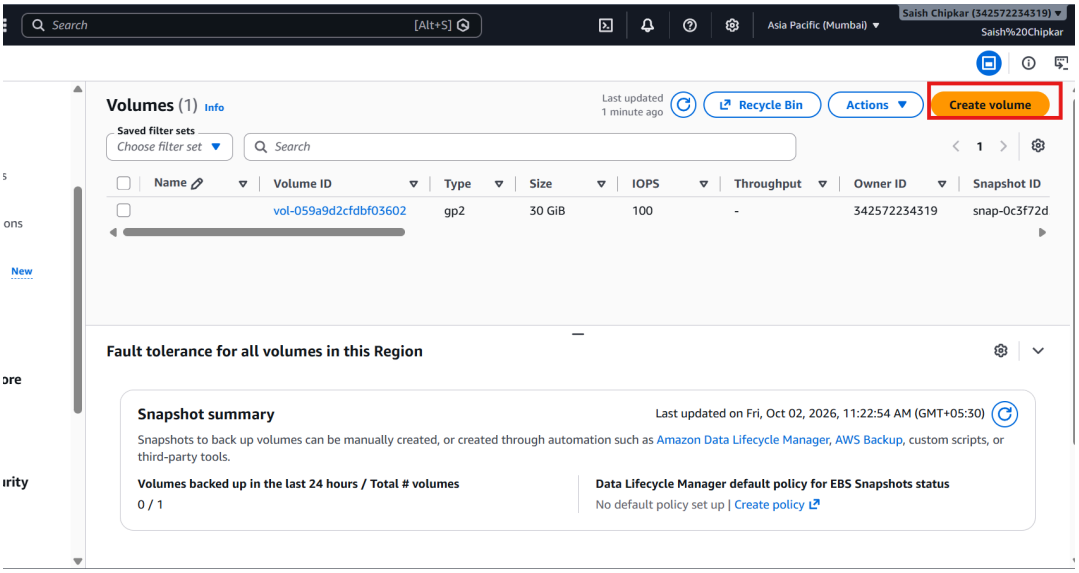
Step 7: Add another group.

Create another group named ebs-volume-disk.
Add the IAM users “Experience1” and “Fresher1”



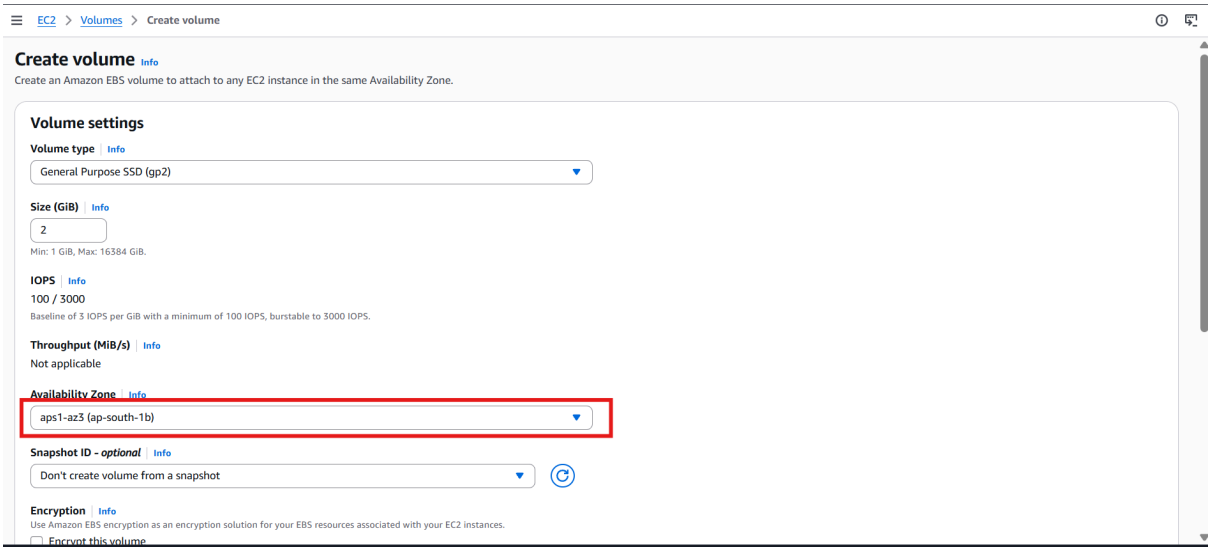
Both of these IAM users can now add volume to the instances.

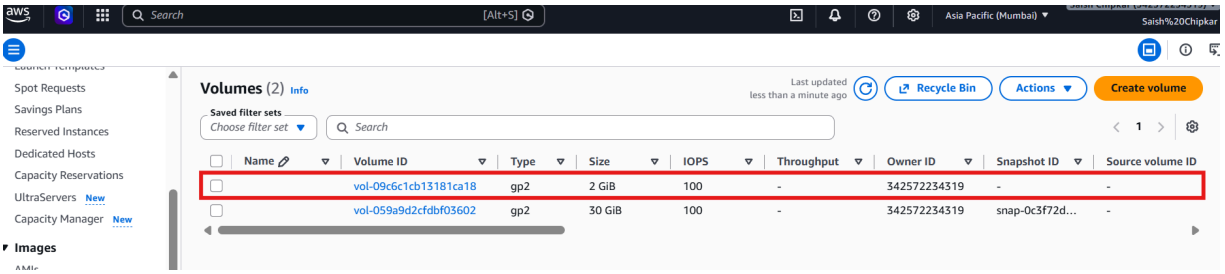
Step 8: Create a volume that is intended to be attached to the Instance.



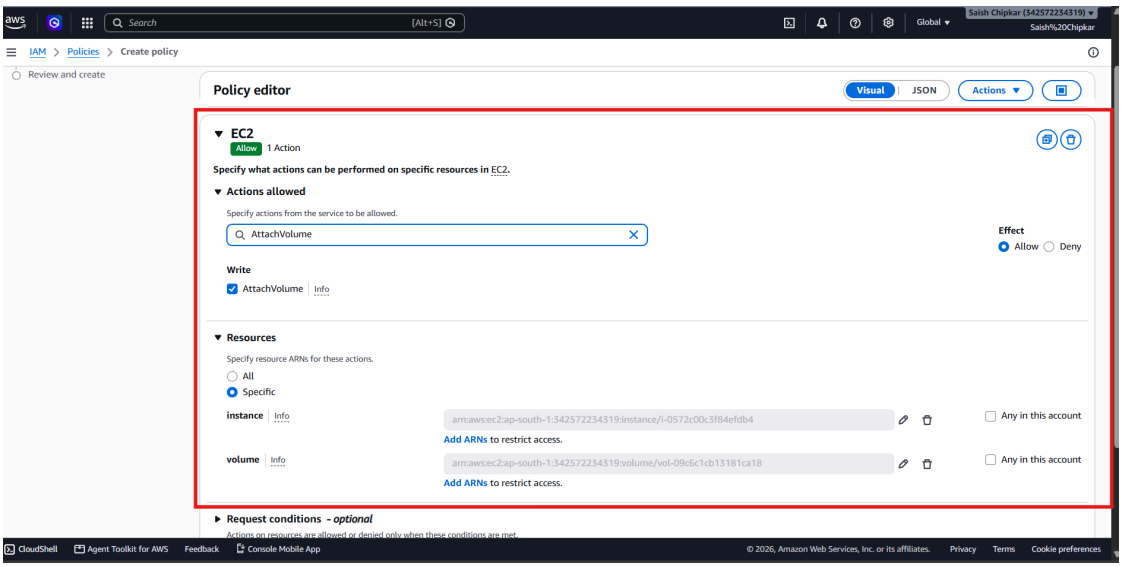
An EBS volume is AZ-specific, so it must be in the same Availability Zone as the EC2 instance to attach it.

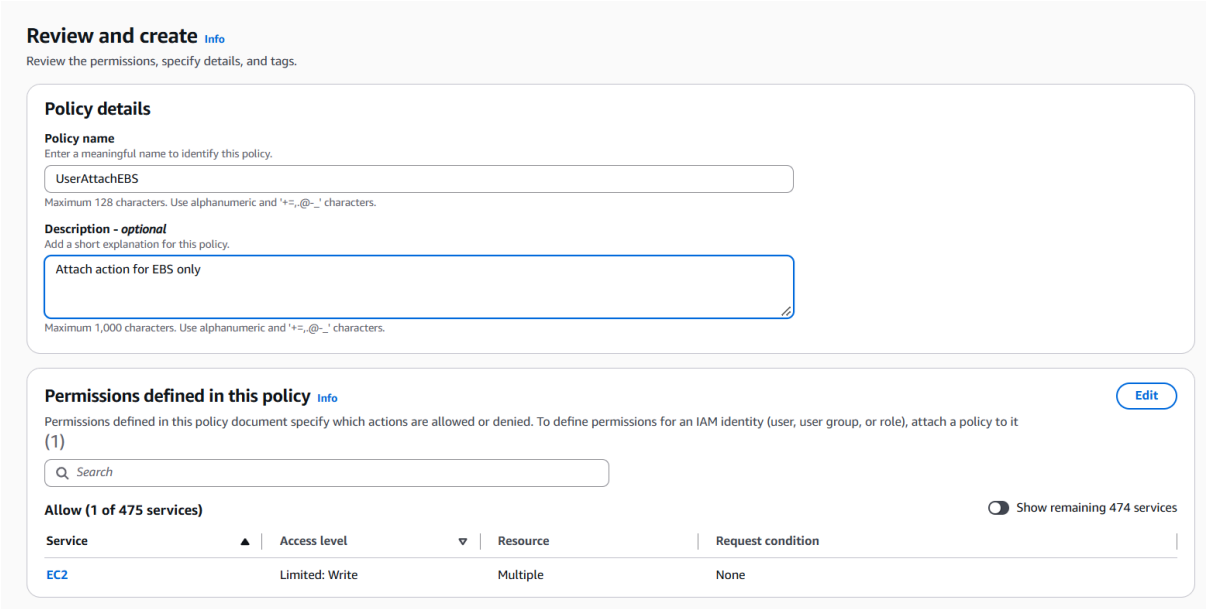
My instance is at ap-south-1b



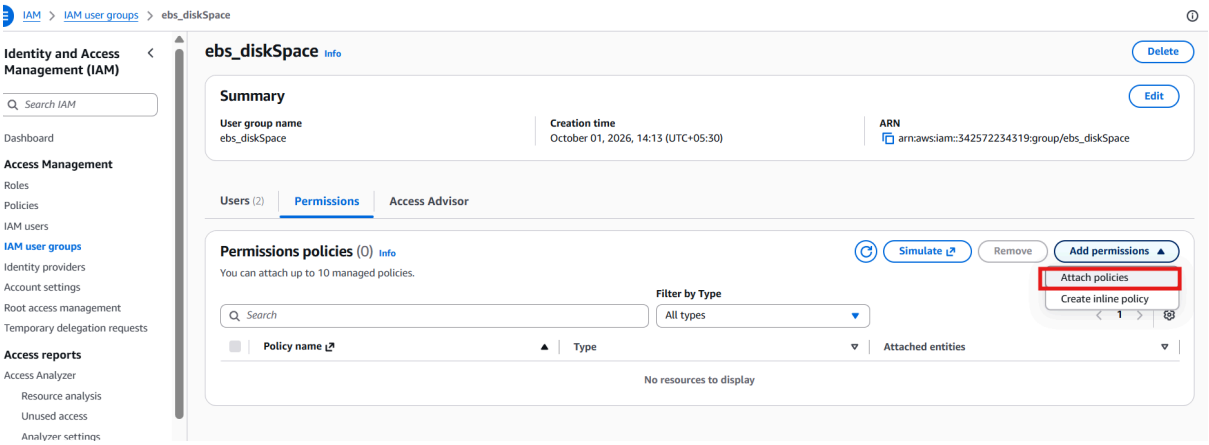


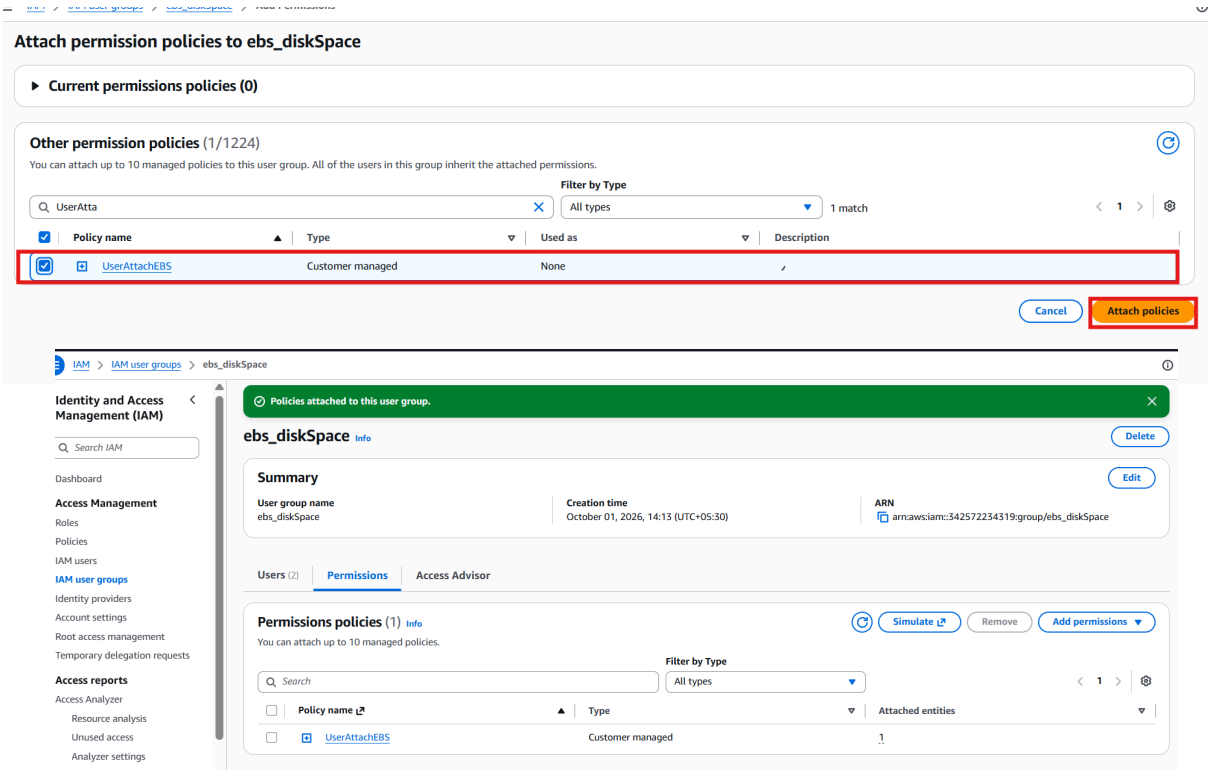
Step 9: Create a User defined policy which allows “AttachVolume” action only for selected Instance and Volume.
Add the Appropriate ARNs for both instance and the required volume.



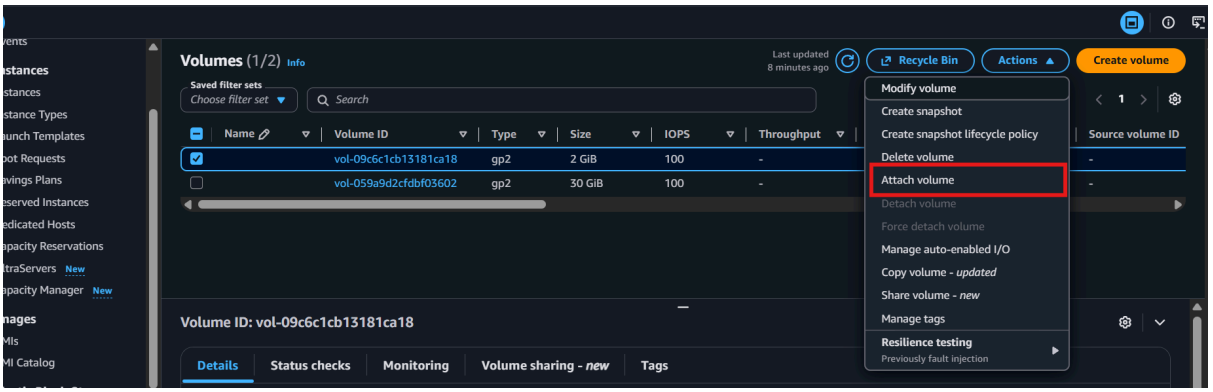


Step 10: Attach the Newly created “UserAttachEBS” to the “ebs_diskSpace” group.

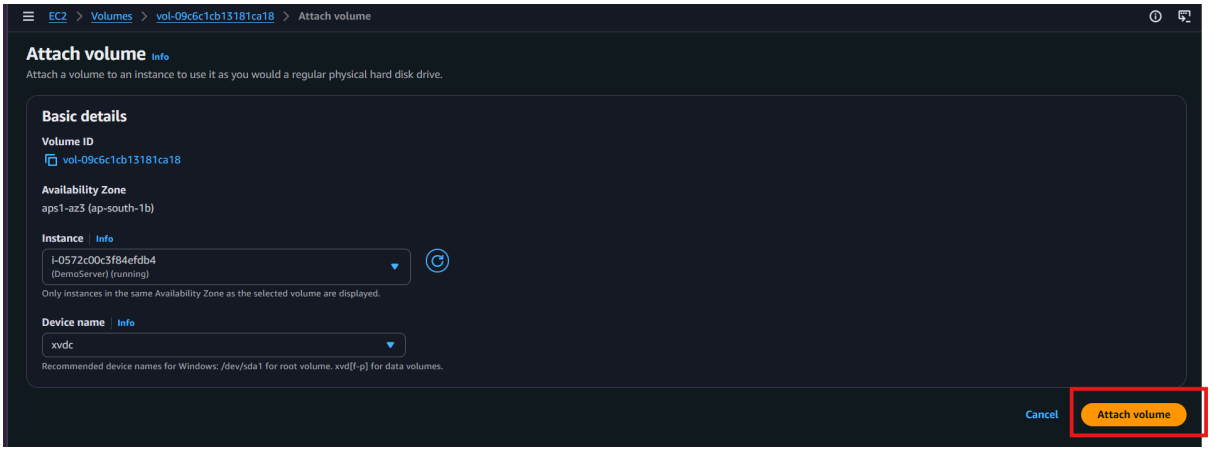


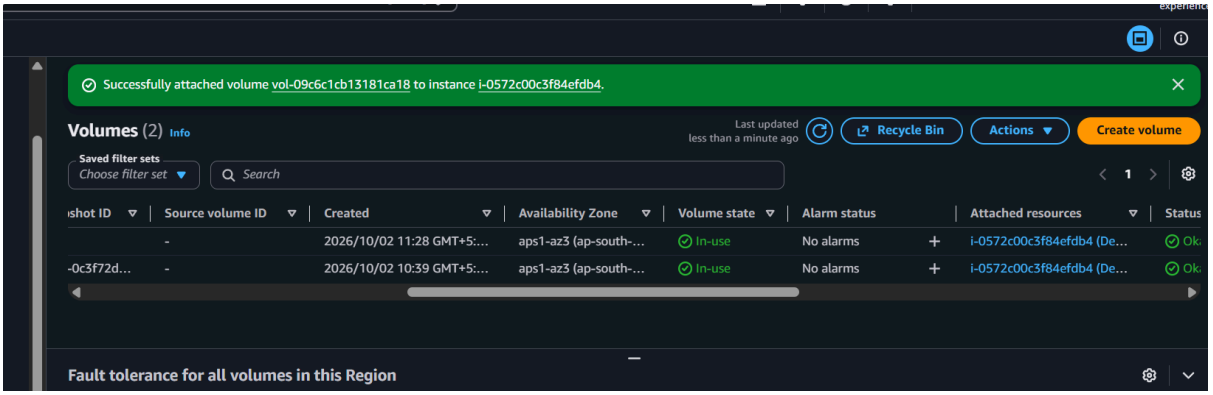


Step 11: Go to the “experienced1” ’s console and attach the volume to the instance.
Note: The IAM user "experienced1" is able to see the Volumes list to because **`AmazonEC2ReadOnlyAccess`** → 👀 allows viewing EC2 resources, including EBS volumes.

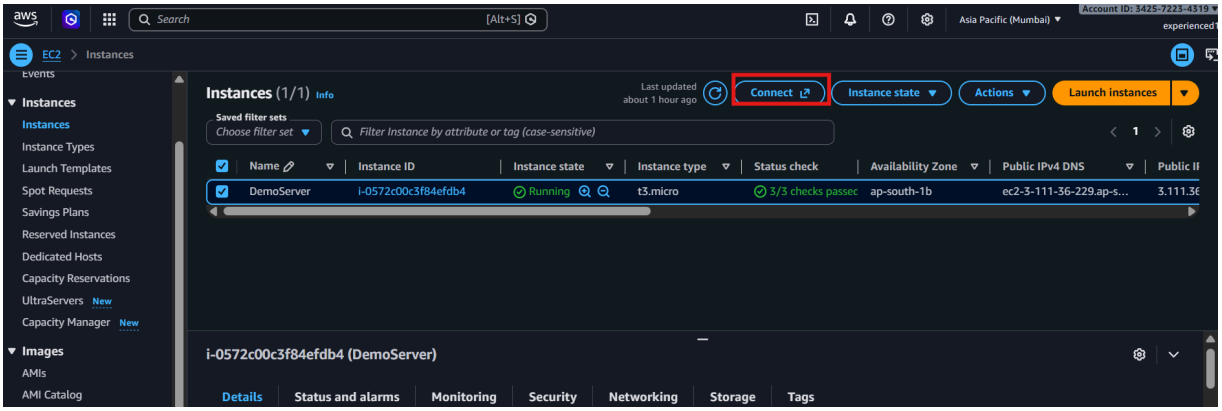


Step 12: Select the required instance name and also the device name. Attach volume

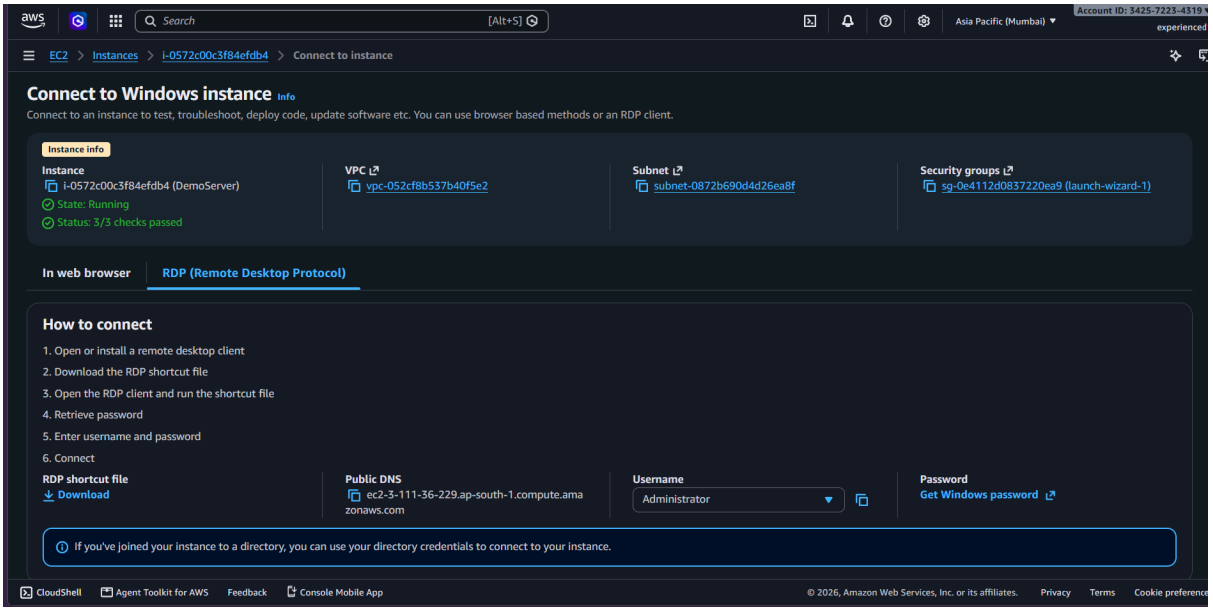




Step 13: Connect to the Instance from the "experienced1" ‘s portal to access the UI.



Connect using RDP.

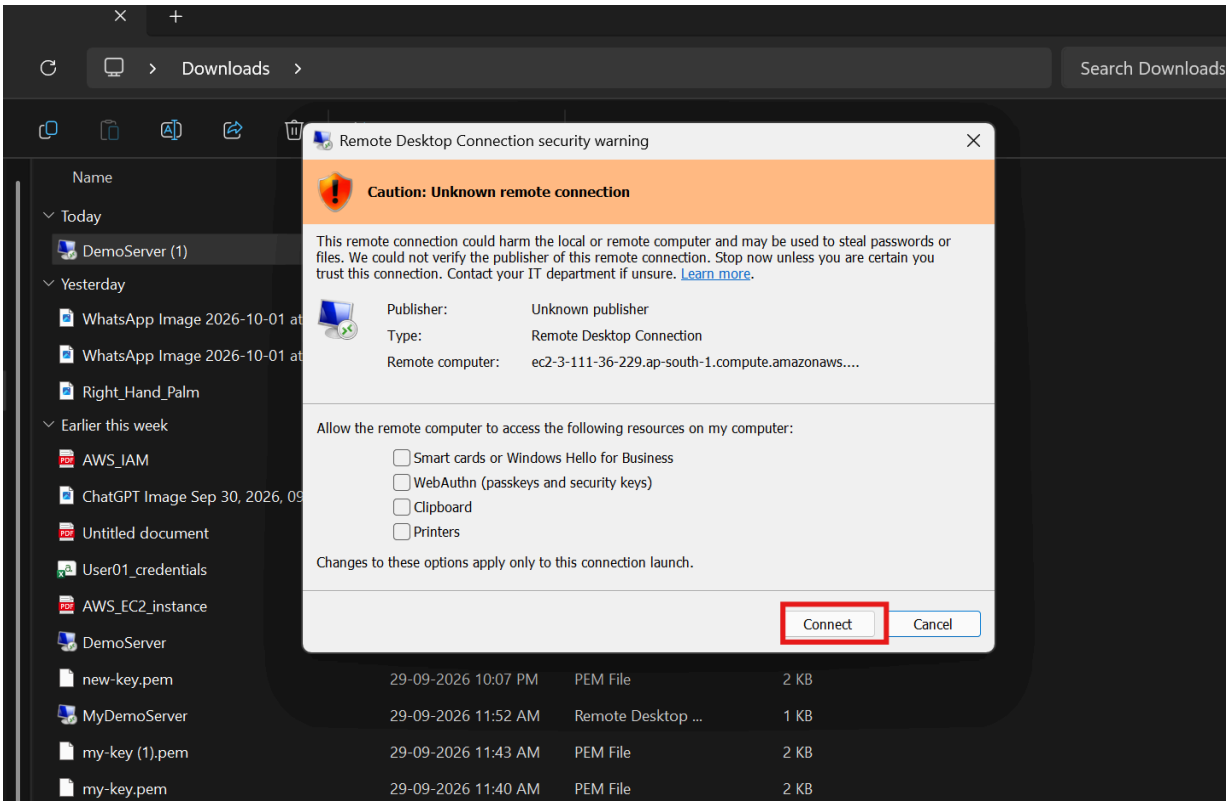


Download the .rdp file.
Note: `.rdp` file is specific to that particular EC2 instance's connection details

It contains connection information such as:

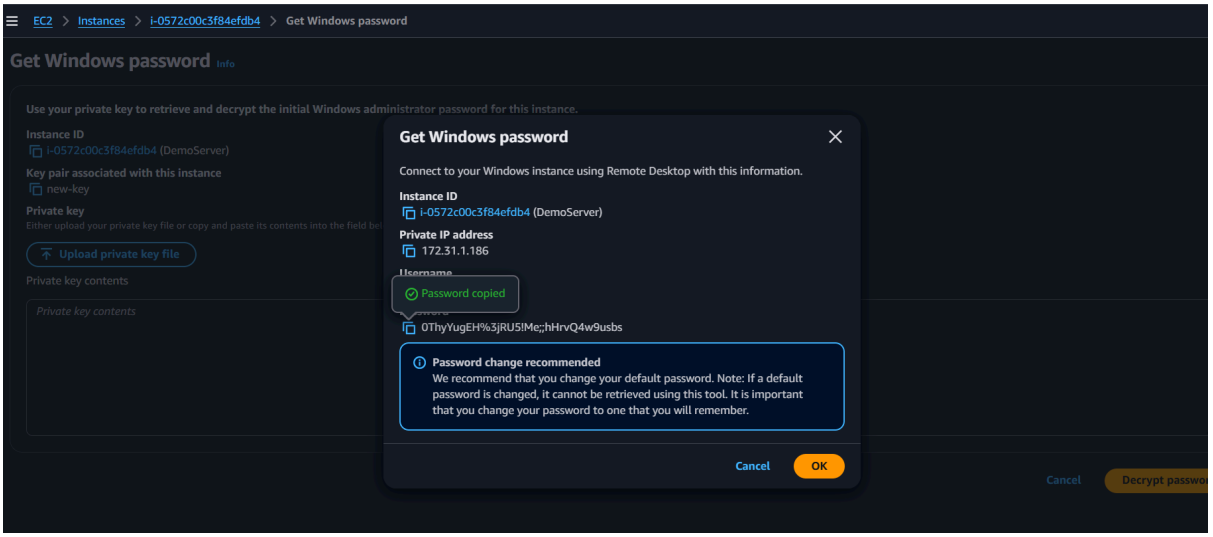
- 🌐 **EC2 instance address/DNS name**
- 🔌 **RDP port** (normally 3389)
- 🖥️ **Remote Desktop settings**
- **Other connection parameters**

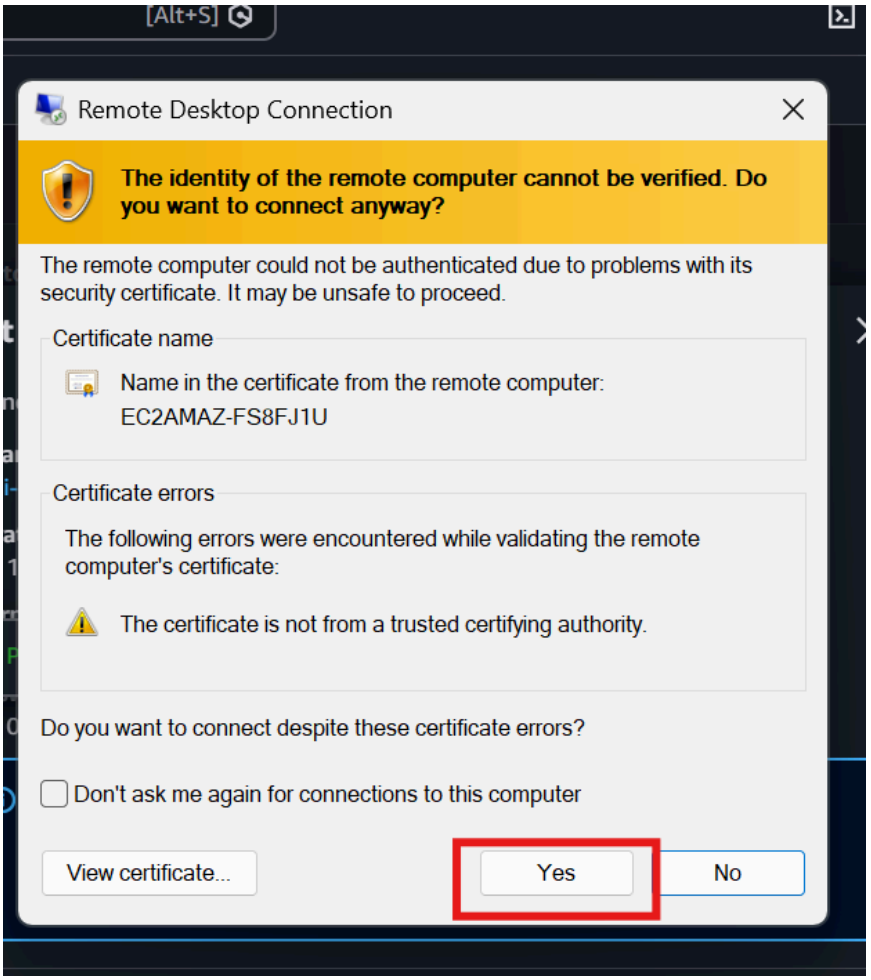
Run the .rdp file



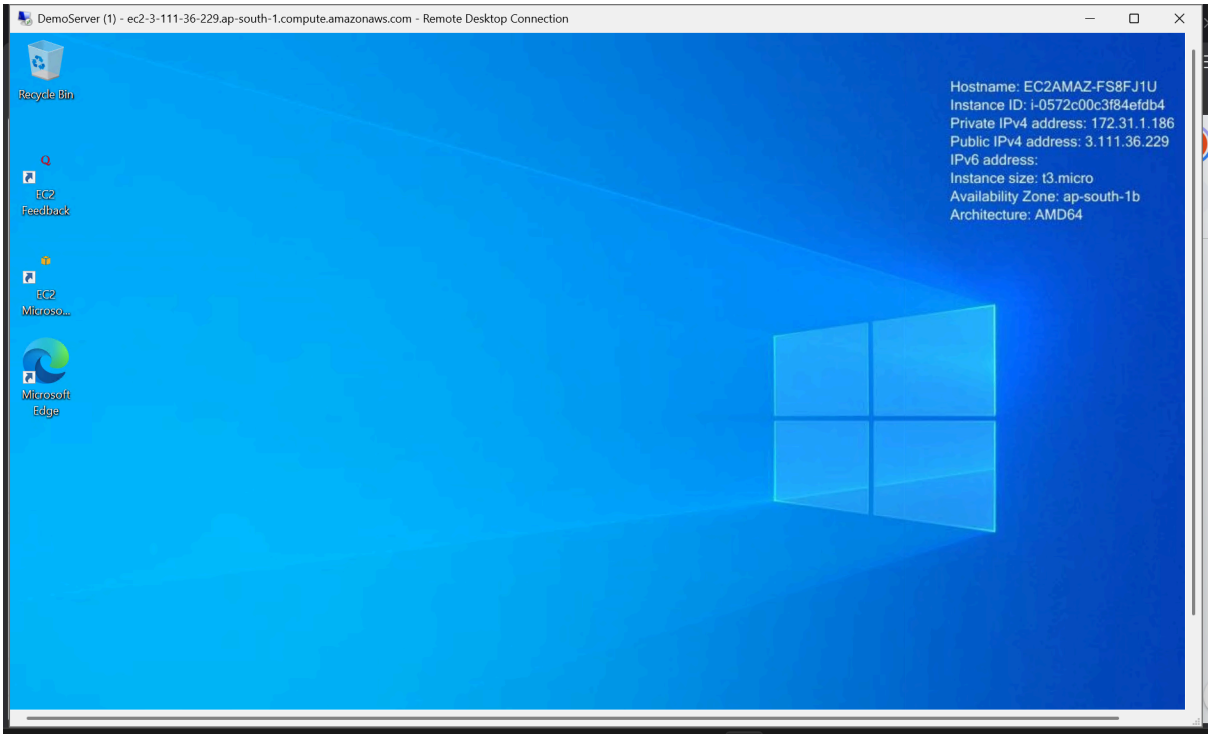
Note: Get windows password form the IAM users account was not allowed. AWS specifically states that the IAM user needs `ec2:GetPasswordData` to retrieve the Windows administrator password.

So I added GetPasswordData action to the User defined policy ->“DemoPolicy” to be able to do that!!

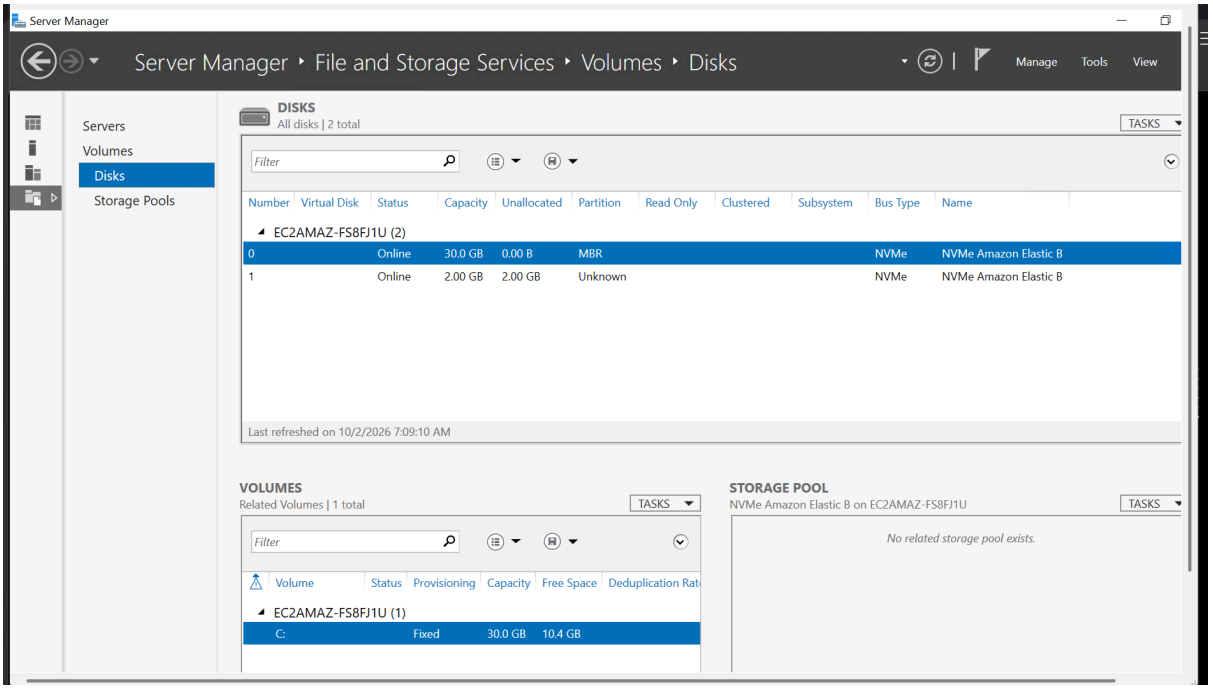




Open the window.

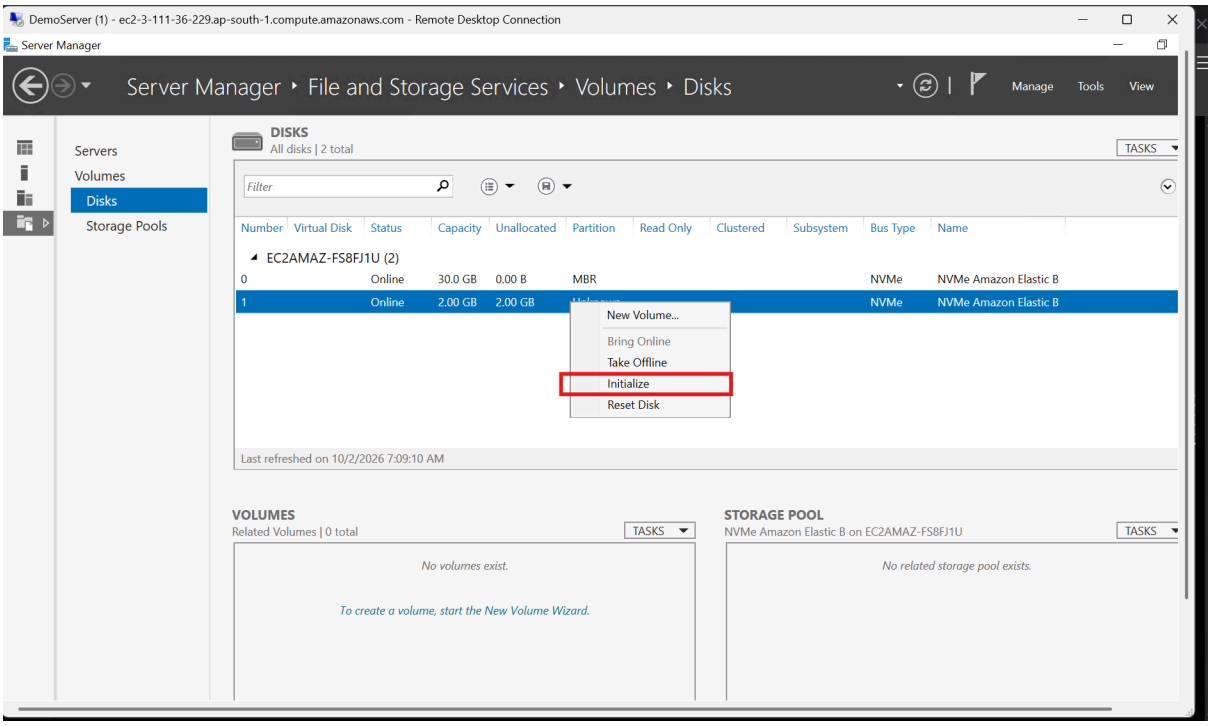


Step 14: Inside of the Manage servers. We see the File and storage option. We see our sees the EBS volume, but **it hasn't been initialized/partitioned yet**.

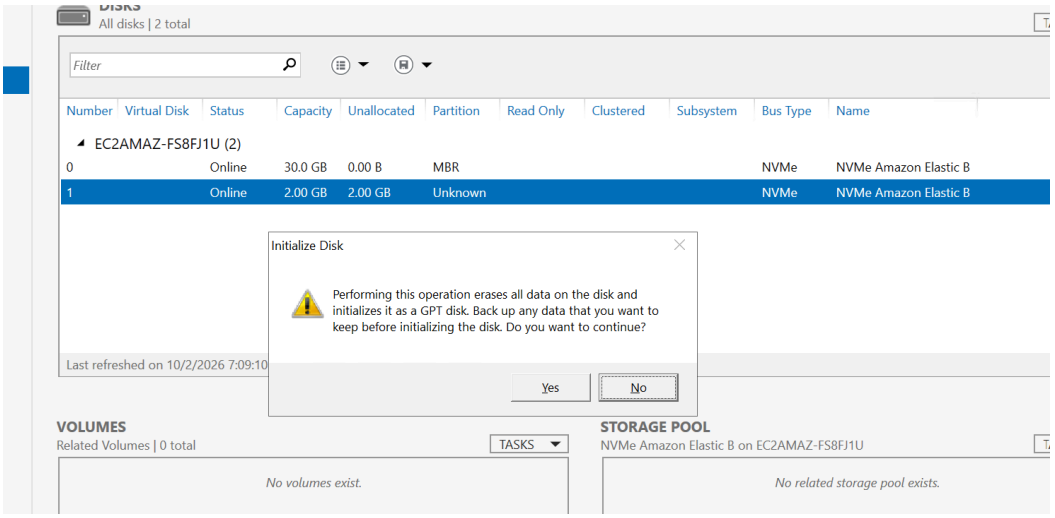


Step 15: Right click on the disk to initialize it.

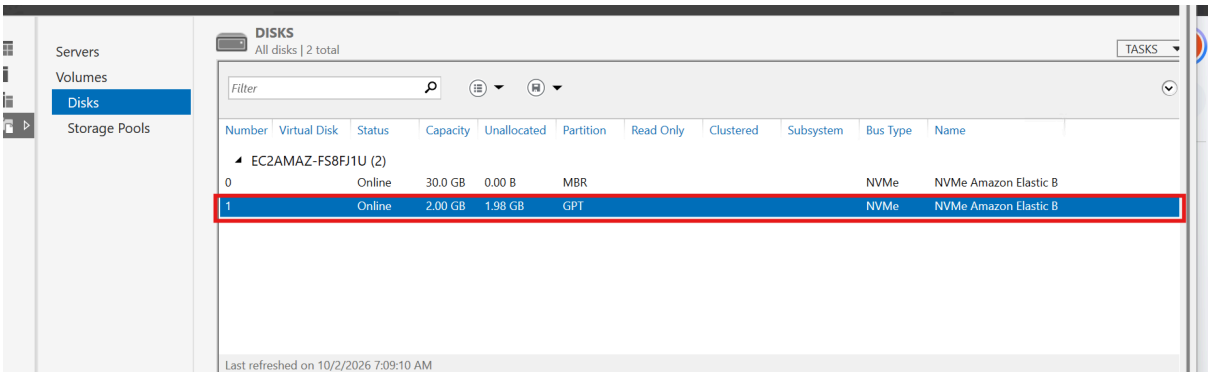
Initialize: Initializing a disk means **preparing a newly attached disk so that Windows can understand and use its storage**. During initialization, Windows creates a **partition table**, which contains information about how the disk's storage will be organized into partitions. In your case, Windows is offering to initialize the 2 GB EBS volume using **GPT (GUID Partition Table)**.



Warning: The warning appears because **initializing can erase existing partition information and make existing data inaccessible**. Therefore, you should initialize a **new, empty EBS volume**, but if the EBS volume already contains important data, you should **not initialize it**. Simply attaching an existing EBS volume to another instance does **not** require initialization.



GPT: GUID Partition Table is a modern standard used to organize and keep track of partitions on a disk. Think of it as the **table of contents for the disk**—it tells Windows where the partitions are and how the disk is organized.



The **2.00 GB** is the total capacity of the EBS volume, but a small amount of space is used/reserved for the **partition table and disk metadata** when you initialize it as GPT.

So approximately:

2.00 GB total − small overhead ≈ 1.98 GB available/unallocated

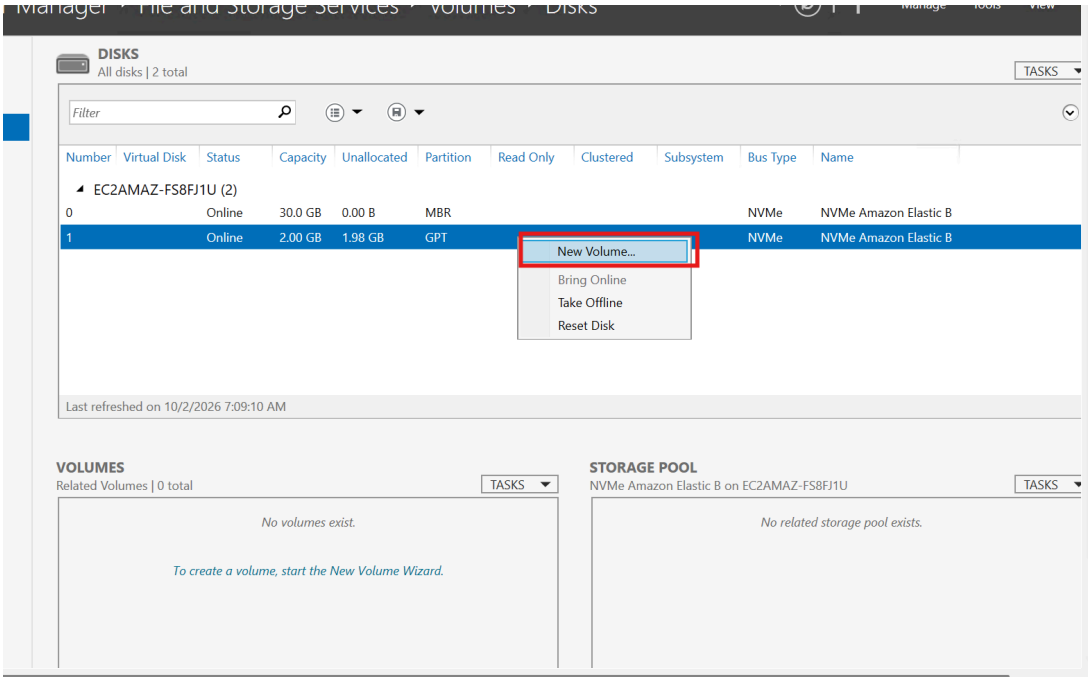
Is the whole memory now one partition?

No, not yet. 👇

- **Capacity:** 2.00 GB → entire EBS disk
- **Partition:** GPT → the disk has been initialized using GPT
- **Unallocated:** 1.98 GB → this space is **not yet a usable partition**

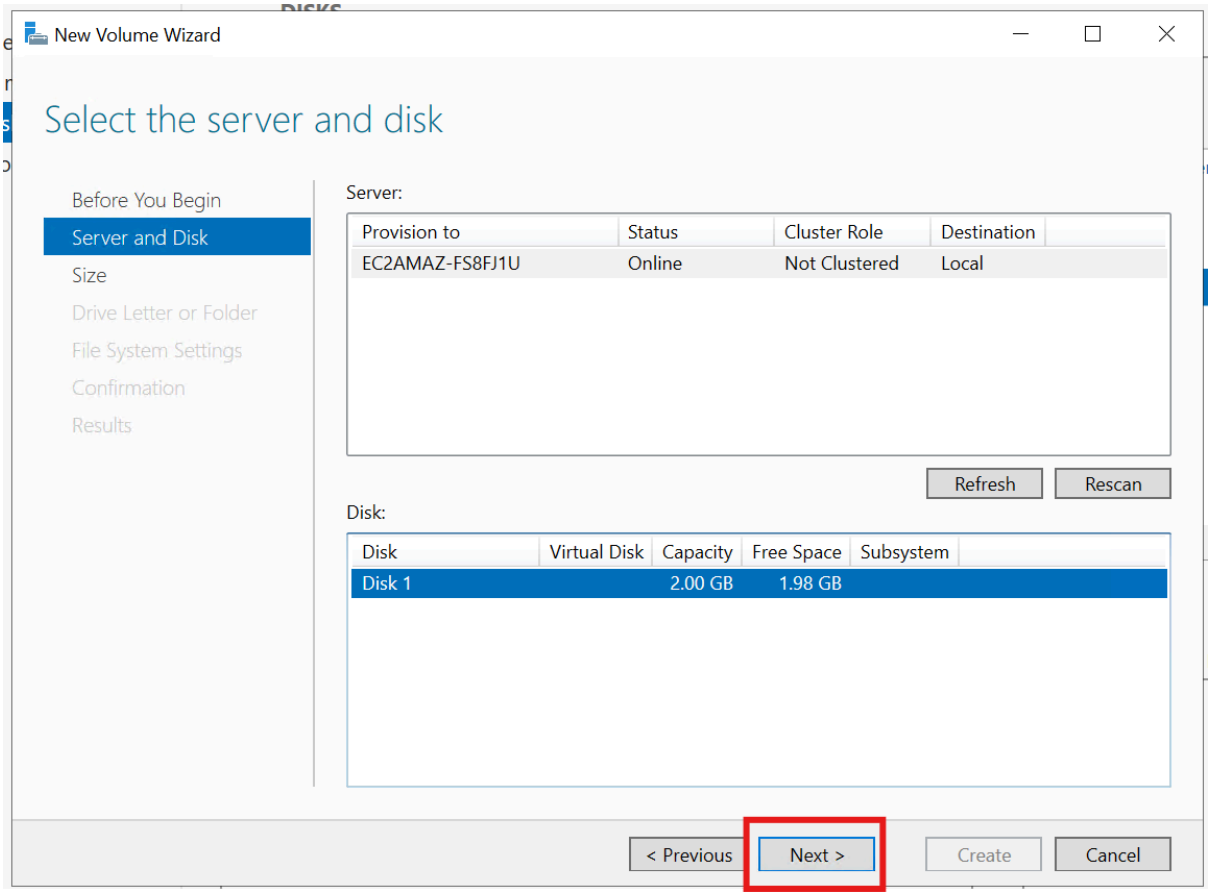
Step 16: After initializing with the GPT. We move with creation of the Partition/Volume.

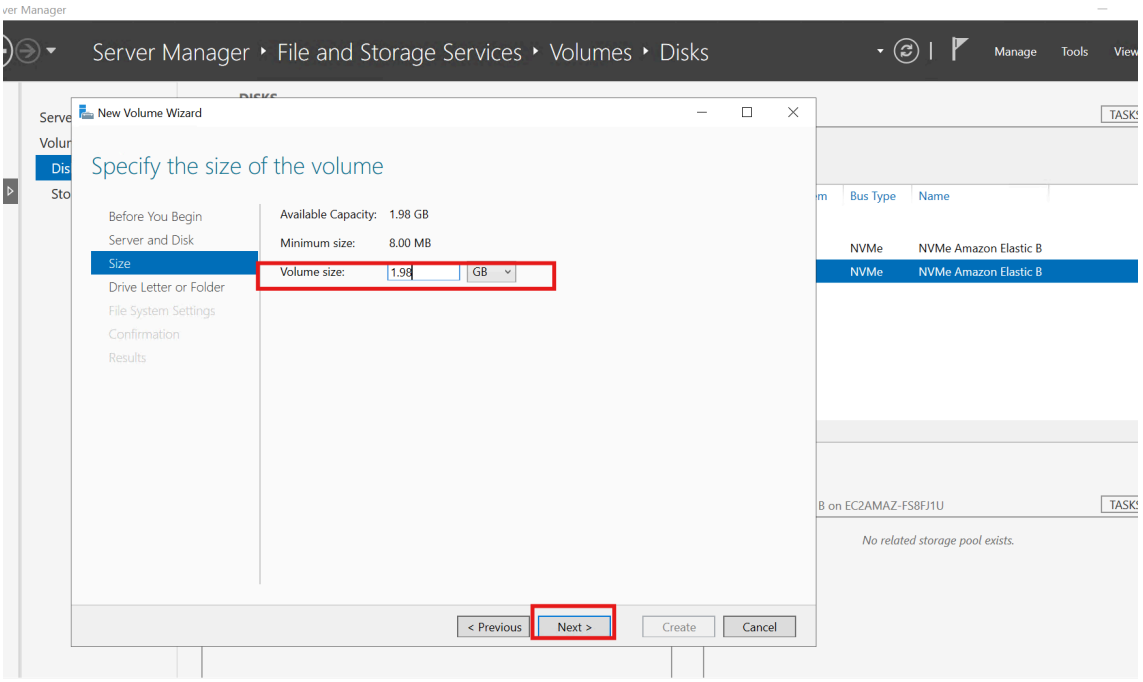
Select new volume



In the **New Volume Wizard**, select your **2 GB Disk 1**.

Use the **maximum available size** if you want the entire ~1.98 GB as one partition.



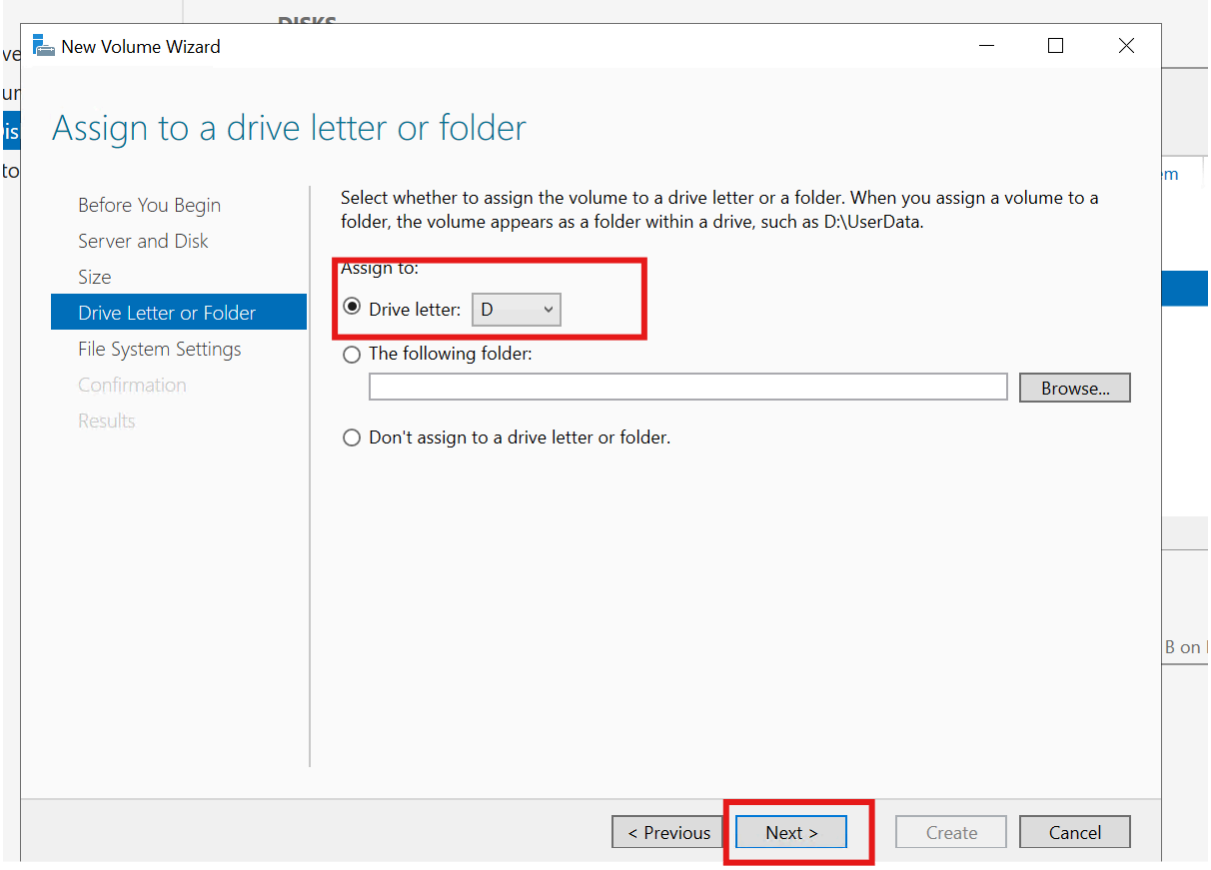


Assign a drive letter, for example **D**

Drive letter → D: ✅ Choose this for your case. You can access the storage as D:\.

The following folder: The storage would appear inside an existing folder, such as C:\UserData, instead of having its own drive letter.

Don't assign: The partition is created but you won't have a normal path to access it yet.

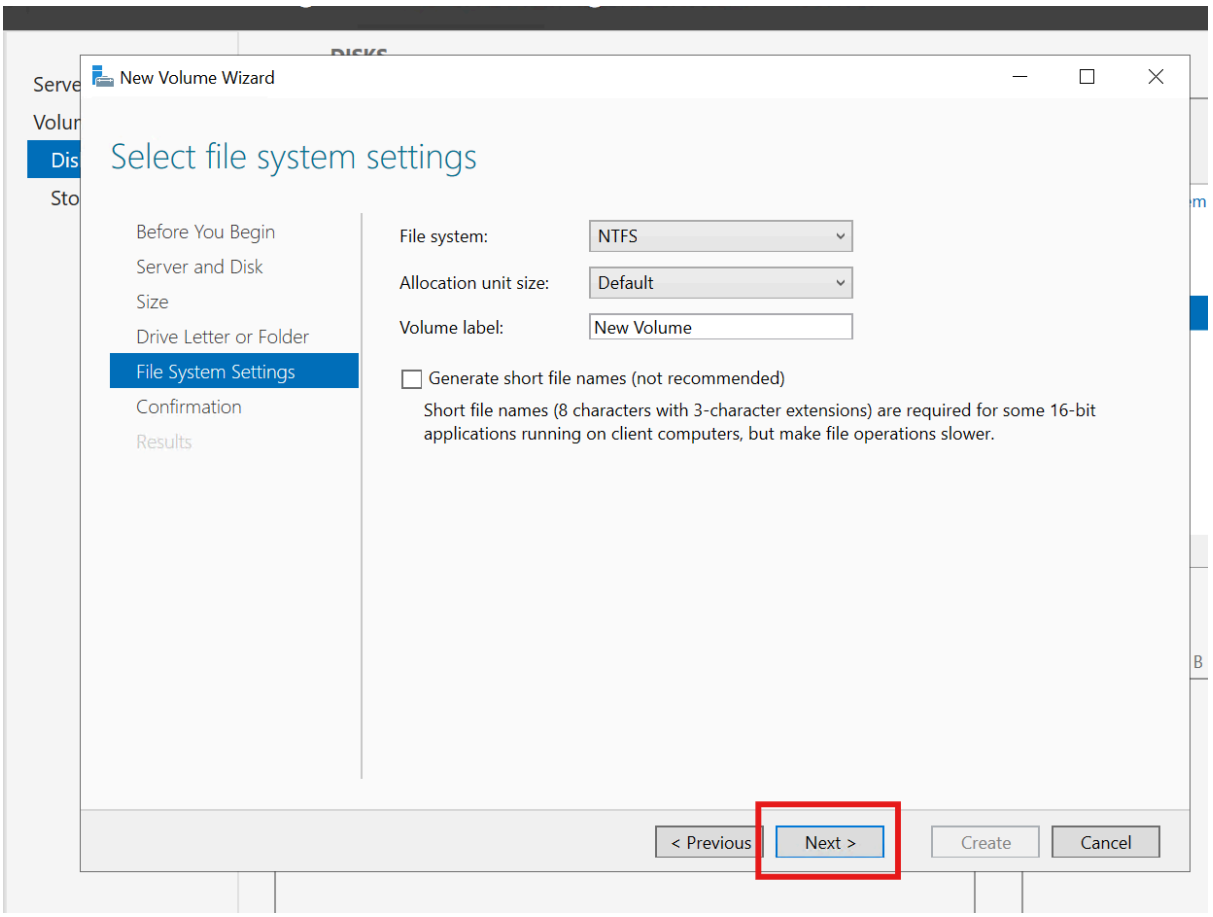


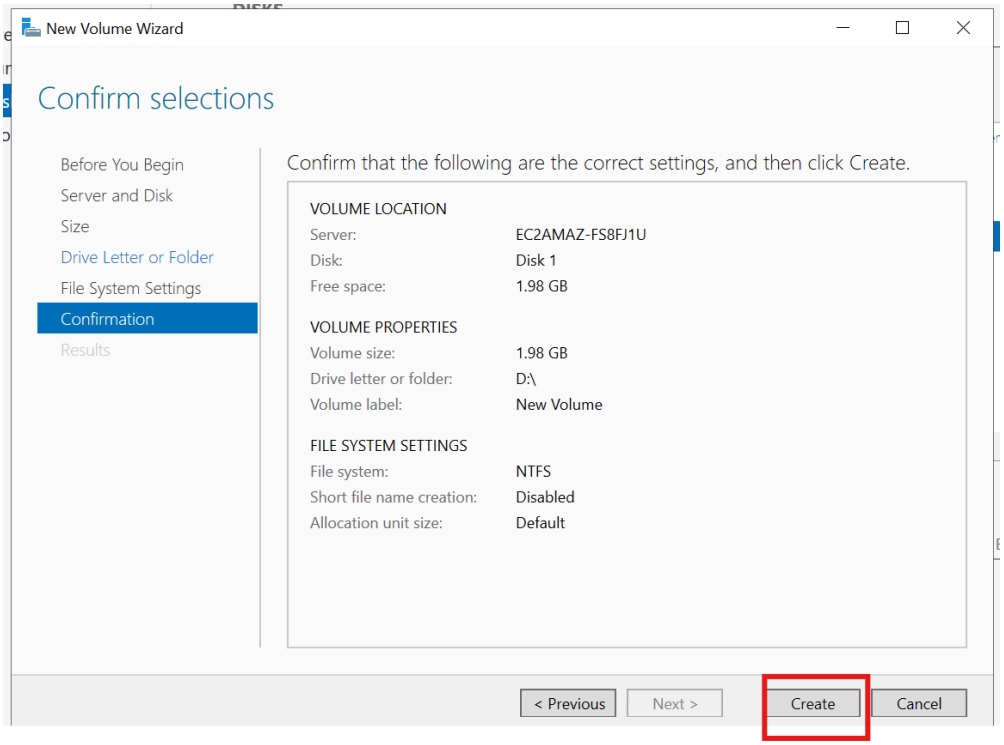
File system: **NTFS** ✅ — the standard Windows file system; it lets Windows store and manage files on the new `D:` drive.

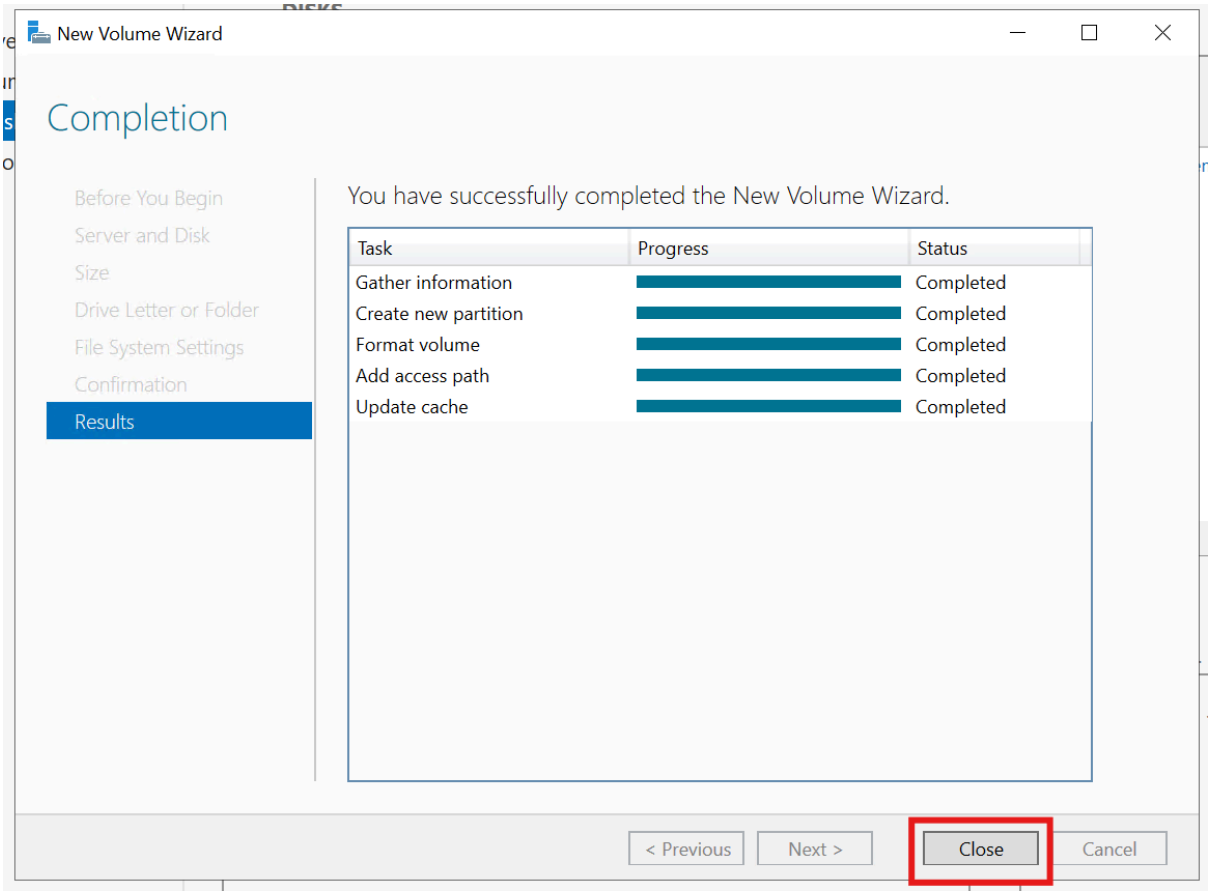
Allocation unit size: **`Default`** ✅ — let Windows choose the appropriate size for storing file blocks.

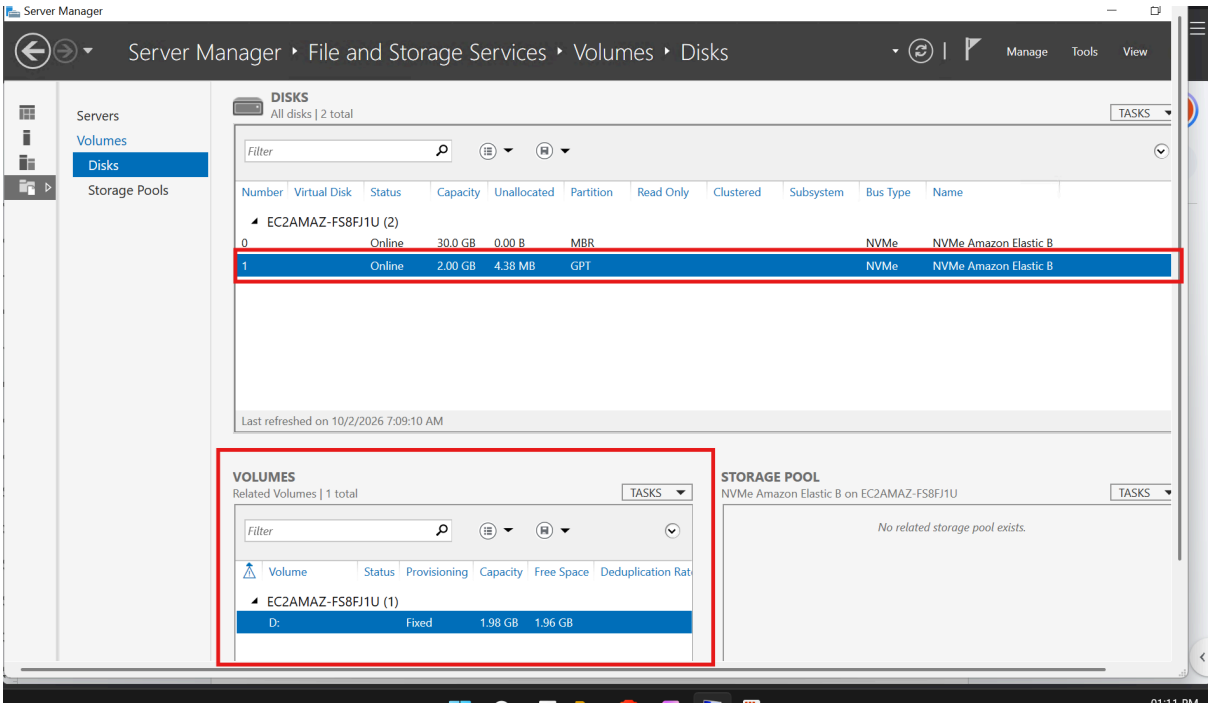
Volume label: **`New Volume`** — optional. You can change it to something meaningful like **`EBS-Data`**.

Generate short file names: Leave **unchecked** ❌ — this is for compatibility with very old applications and isn't needed here.

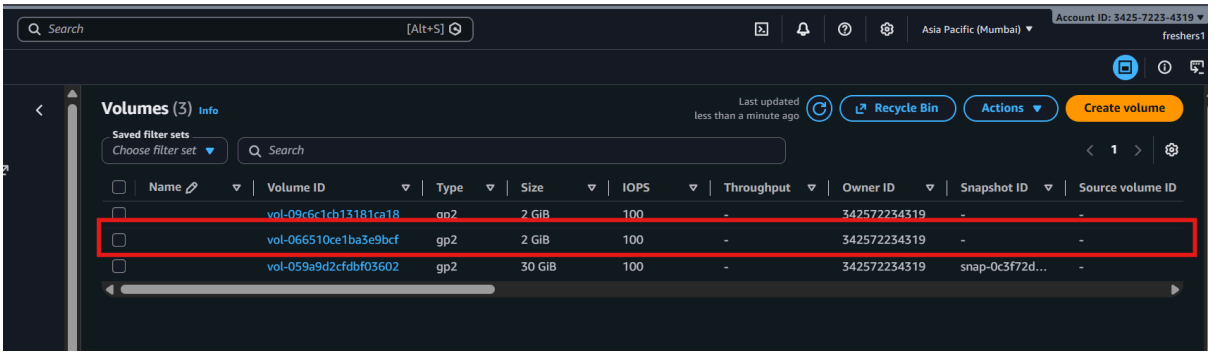

New Volume Wizard
Select file system settings
Before You Begin
Server and Disk
Size
Drive Letter or Folder
File System Settings
Confirmation
Results
File system:
NTFS
Allocation unit size:
Default
Volume label:
New Volume
Generate short file names (not recommended)
Short file names (8 characters with 3-character extensions) are required for some 16-bit applications running on client computers, but make file operations slower.
< Previous
Next >
Create
Cancel

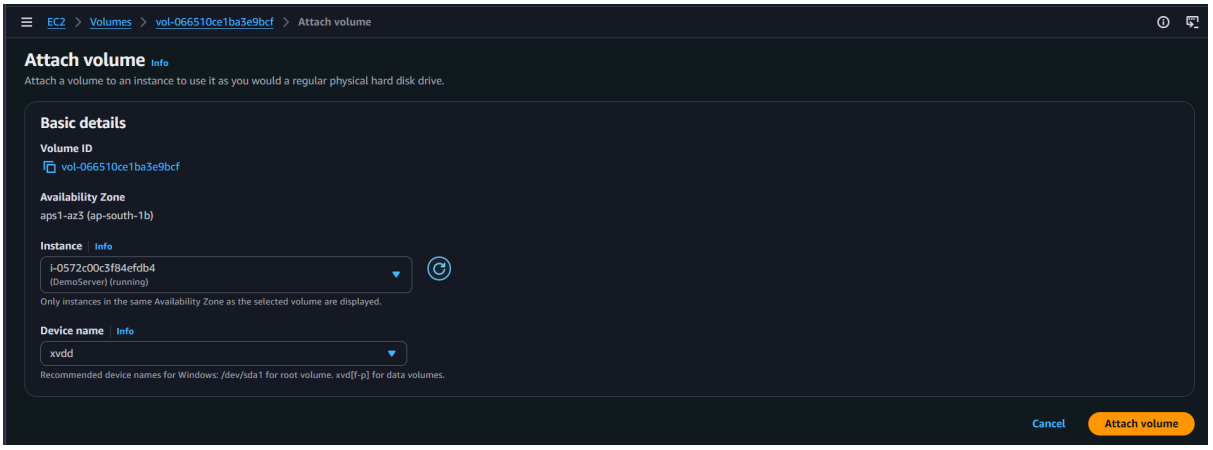

New Volume Wizard
Confirm selections
Before You Begin
Server and Disk
Size
Drive Letter or Folder
File System Settings
Confirmation
Results
Confirm that the following are the correct settings, and then click Create.
VOLUME LOCATION
Server: EC2AMAZ-FS8FJ1U
Disk: Disk 1
Free space: 1.98 GB
VOLUME PROPERTIES
Volume size: 1.98 GB
Drive letter or folder: D:\
Volume label: New Volume
FILE SYSTEM SETTINGS
File system: NTFS
Short file name creation: Disabled
Allocation unit size: Default
< Previous
Next >
Create
Cancel


New Volume Wizard
Completion
Before You Begin
Server and Disk
Size
Drive Letter or Folder
File System Settings
Confirmation
Results
You have successfully completed the New Volume Wizard.
Task
Progress
Status
Gather information
Completed
Create new partition
Completed
Format volume
Completed
Add access path
Completed
Update cache
Completed
< Previous
Next >
Close
Cancel

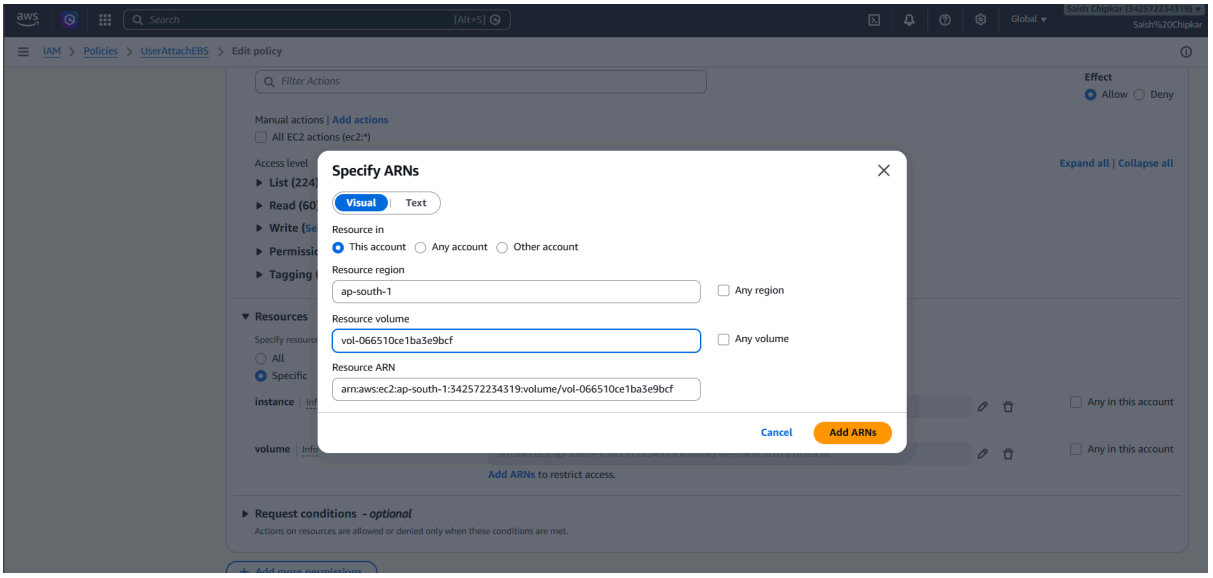

Server Manager
Server Manager ‣ File and Storage Services ‣ Volumes ‣ Disks
Manage
Tools
View
Servers
Volumes
Disks
Storage Pools
DISKS
All disks | 2 total
TASKS
Filter
Number
Virtual Disk
Status
Capacity
Unallocated
Partition
Read Only
Clustered
Subsystem
Bus Type
Name
EC2AMAZ-FS8FJ1U (2)
0
Online
30.0 GB
0.00 B
MBR
NVMe
NVMe Amazon Elastic B
1
Online
2.00 GB
4.38 MB
GPT
NVMe
NVMe Amazon Elastic B
Last refreshed on 10/2/2026 7:09:10 AM
VOLUMES
Related Volumes | 1 total
TASKS
Filter
Volume
Status
Provisioning
Capacity
Free Space
Deduplication Rat
EC2AMAZ-FS8FJ1U (1)
D:
Fixed
1.98 GB
1.96 GB
STORAGE POOL
NVMe Amazon Elastic B on EC2AMAZ-FS8FJ1U
TASKS
No related storage pool exists.

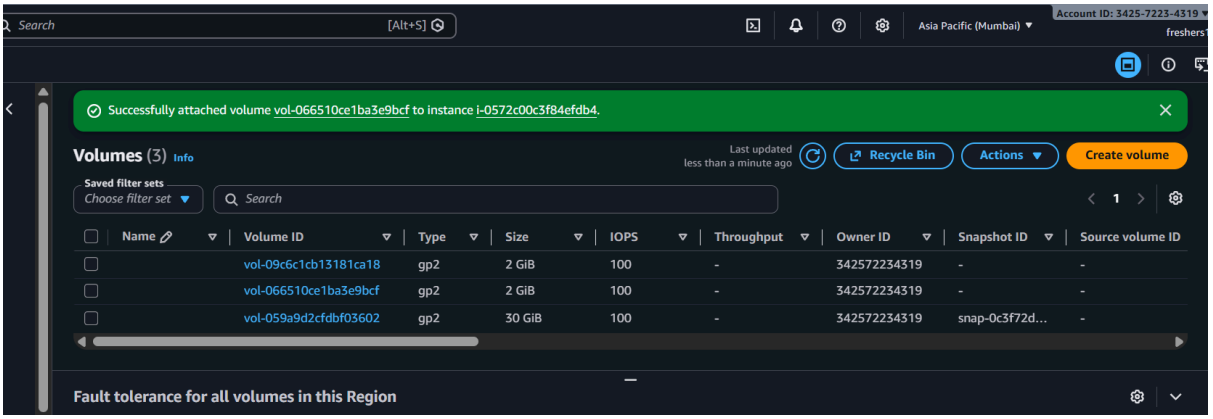
Step 17: Create a EBS from the IAM user “Fresher1” for the same instance.



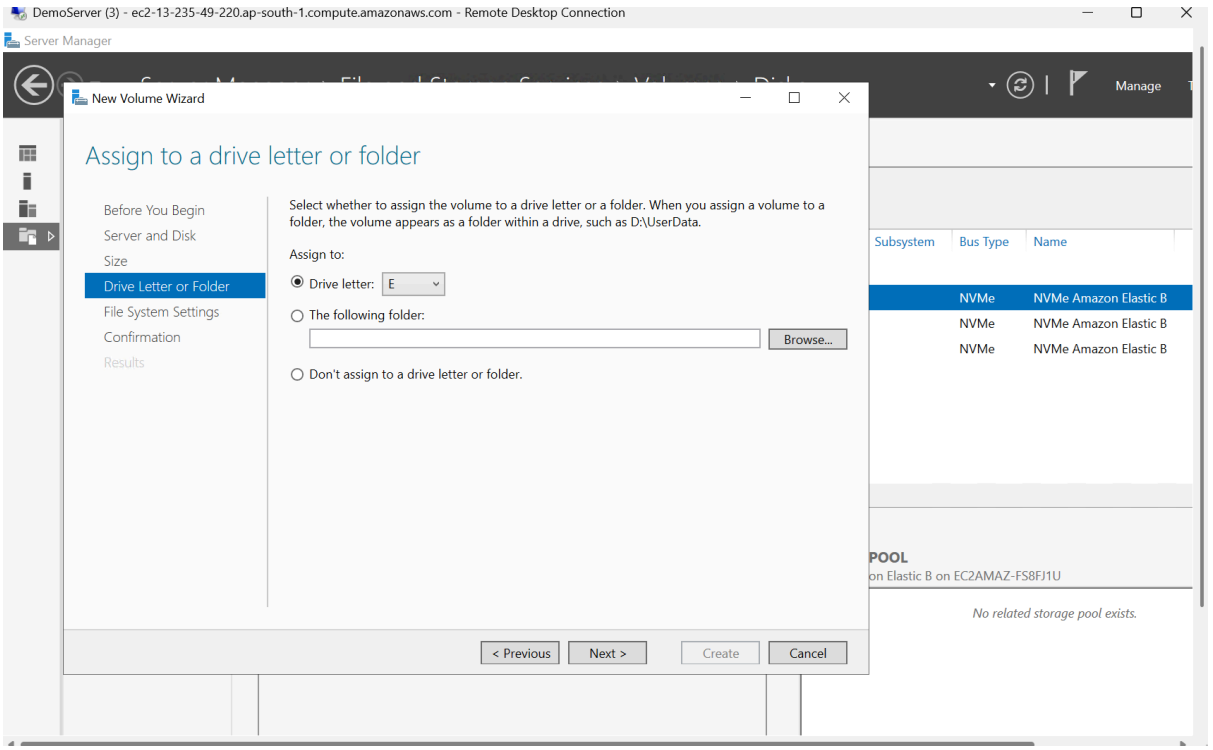


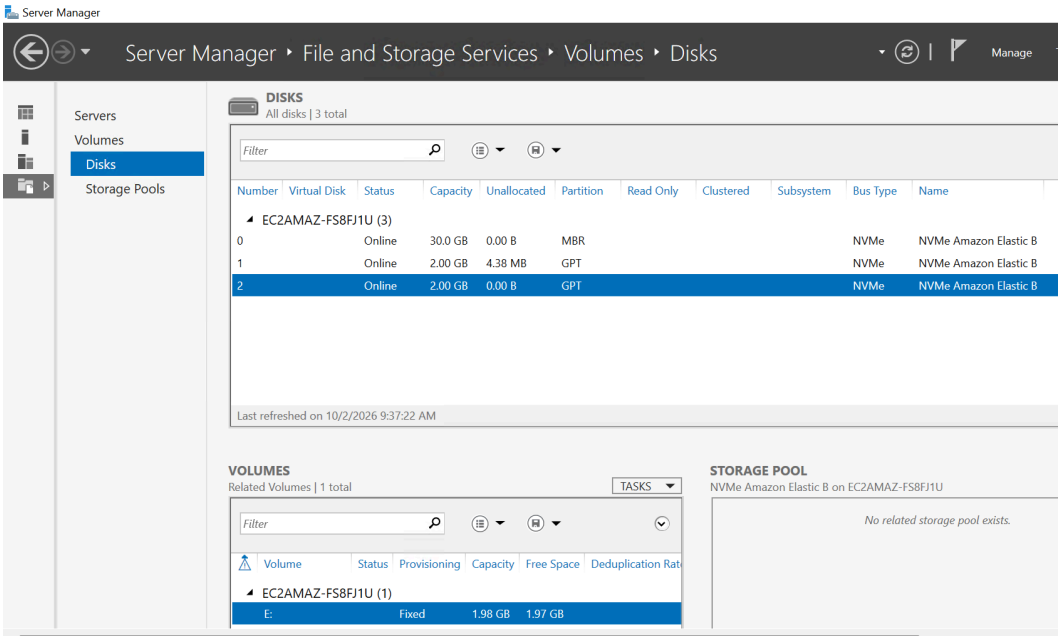
Edit the “UserAttachEBS” policy to include the permission to attach the newly created volume too, by adding the ARN of the volume.





Step 18: Connect to the Instance from the "freshers1" ‘s portal to access the UI.
Repeat form step 13 to 16. To add E drive to the instance.





Topology Diagram:

